

SYSTEM & UML DIAGRAMS

AI Chatbot for College · BBA Computer Applications

1. Use Case Diagram

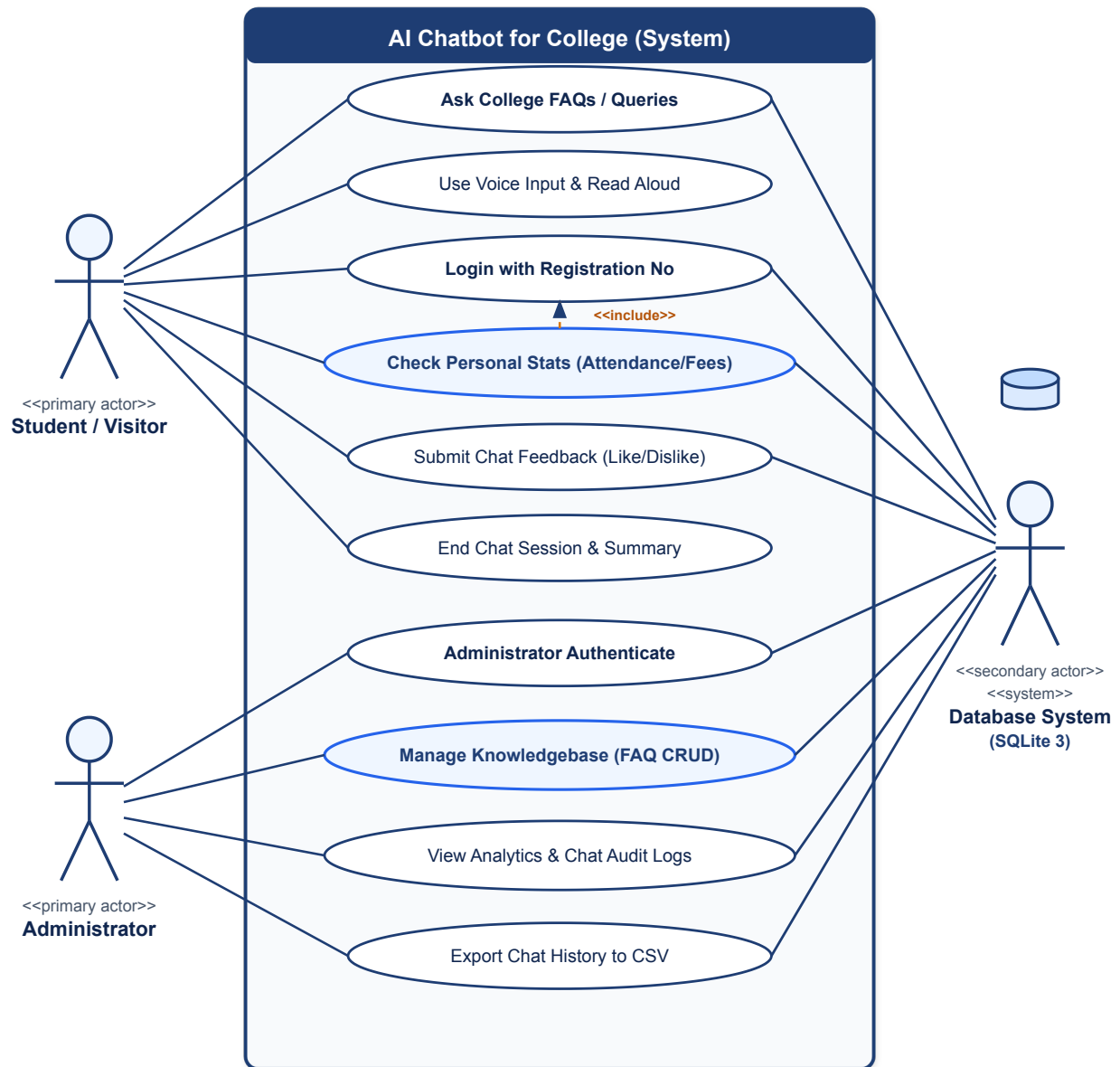


Fig 1: Use Case Diagram — primary actors (Student & Administrator), system boundary, and secondary actor (SQLite Database System) with data-tier associations.

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2. Class Diagram (With Full Relations & Multiplicities)

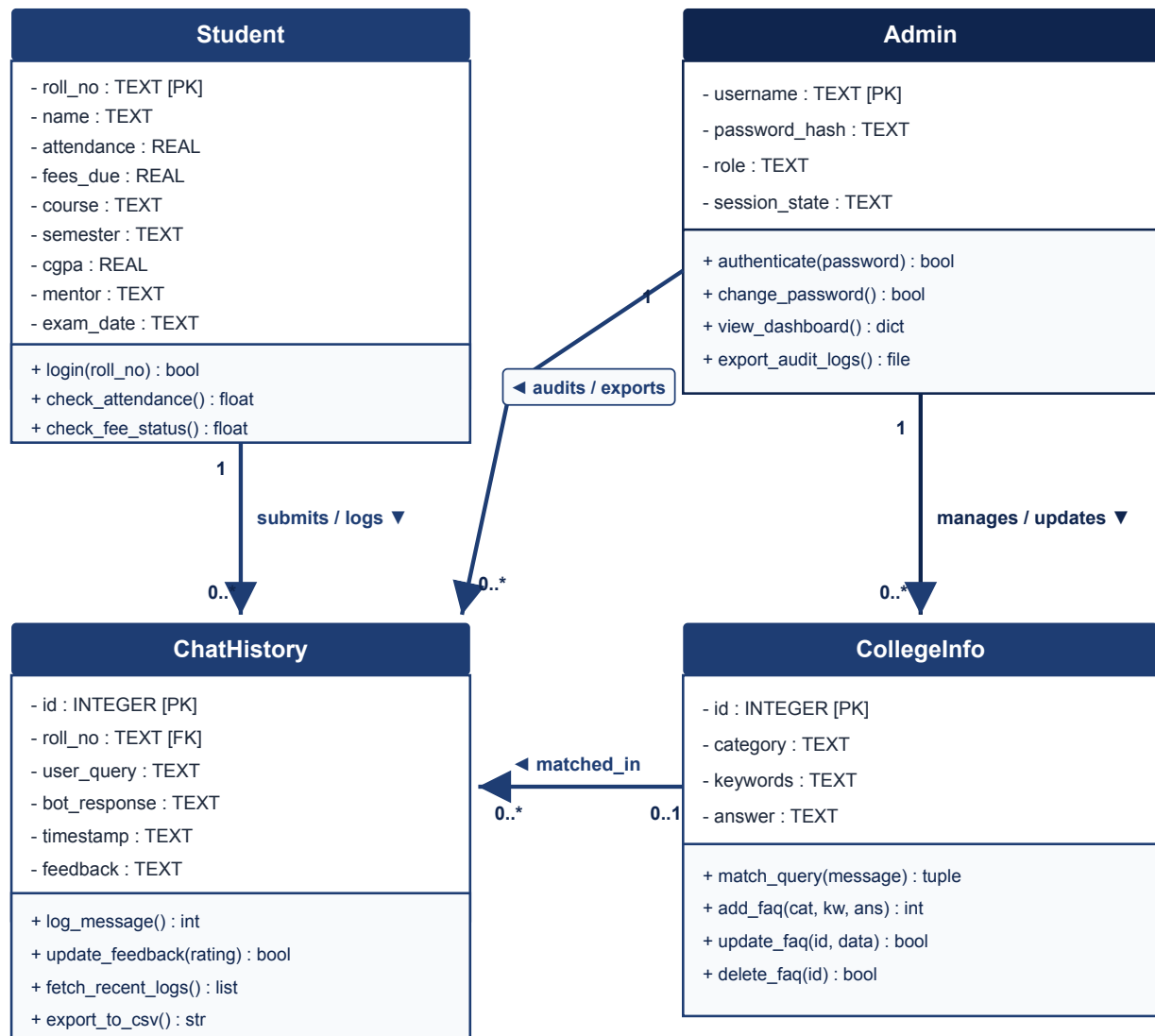


Fig 2: Class Diagram — core entities, attributes, operations, and 100% solid association arrows with multiplicities and role verbs.

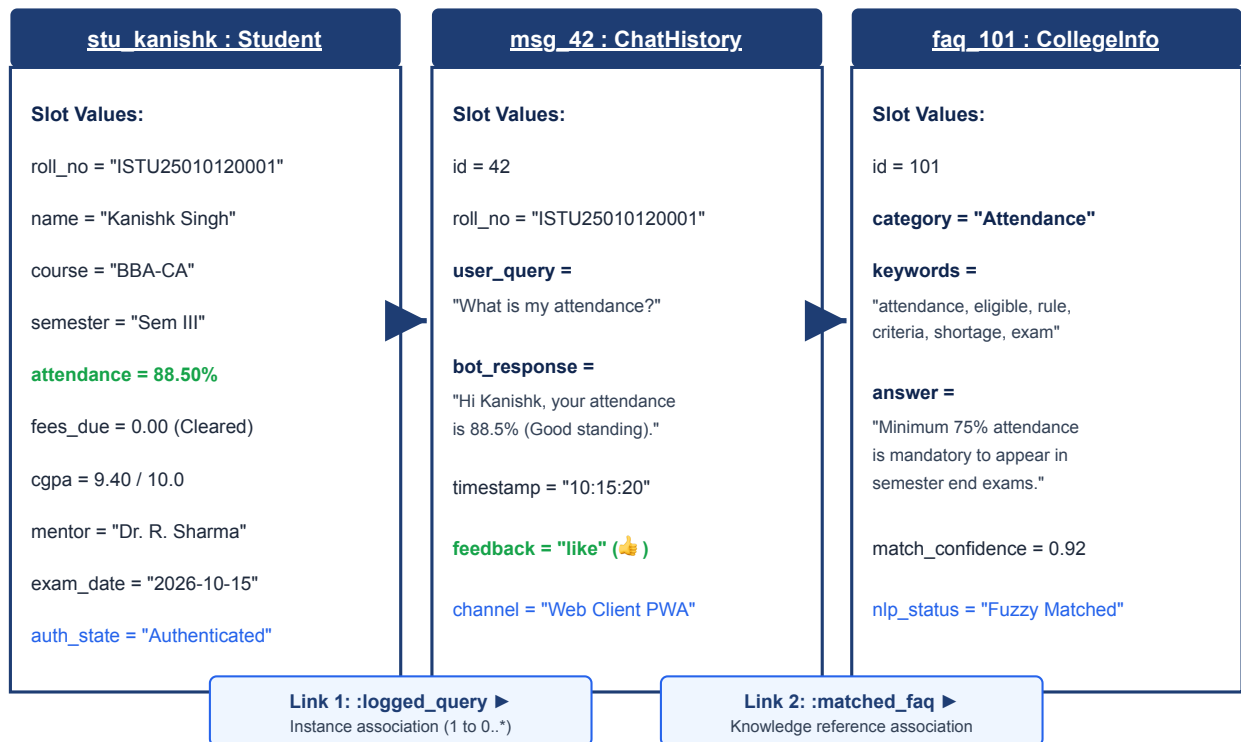
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3. Object Diagram (Runtime Instances & Slot Values)

Live Execution Snapshot Scenario (Active Student Transaction):

Student Kanishk Singh asks about personal attendance; ChatHistory logs query #42 matched against CollegeInfo FAQ #101.



✓ Standard UML Object Diagram Specifications Verified:

- Exactly 3 Concrete Runtime Object Instances: <u>Student</u>, <u>ChatHistory</u>, and <u>CollegeInfo</u>.
- Fully populated slot values with actual live strings, numeric metrics (88.5%, CGPA 9.40), and timestamps.
- 100% Solid Directional Association Arrows between instances (zero dashed lines).

Fig 3: Object Diagram — exactly 3 runtime instances (Student, ChatHistory, CollegeInfo) with concrete slots and solid association arrows.

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4. Sequence Diagram — Query, Feedback & Session Termination

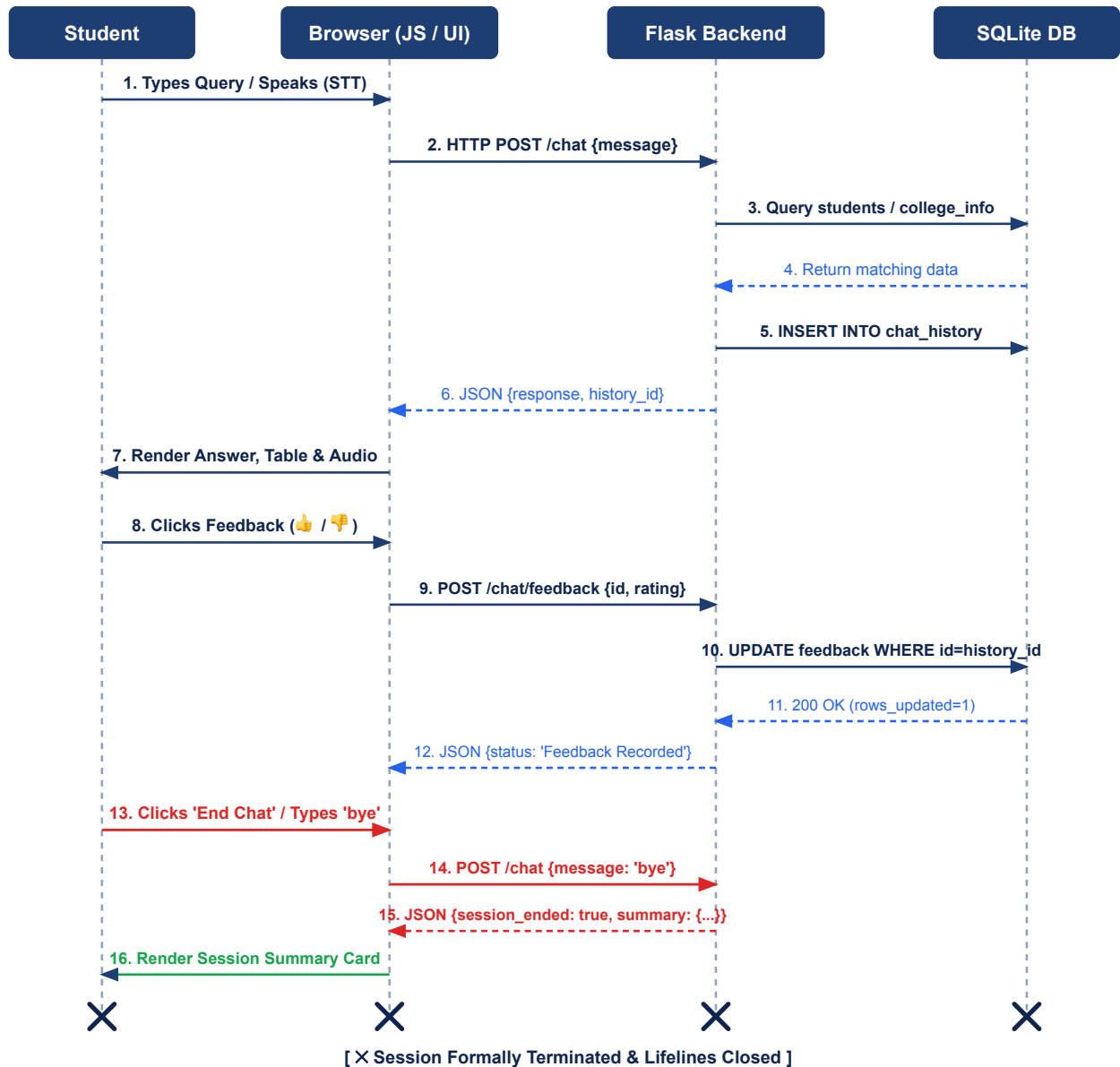


Fig 4: Sequence Diagram — complete query, feedback ACK, explicit End Chat flow, and standard UML lifeline termination (X).

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5. Activity Diagram — Chatbot Response & Session Flow

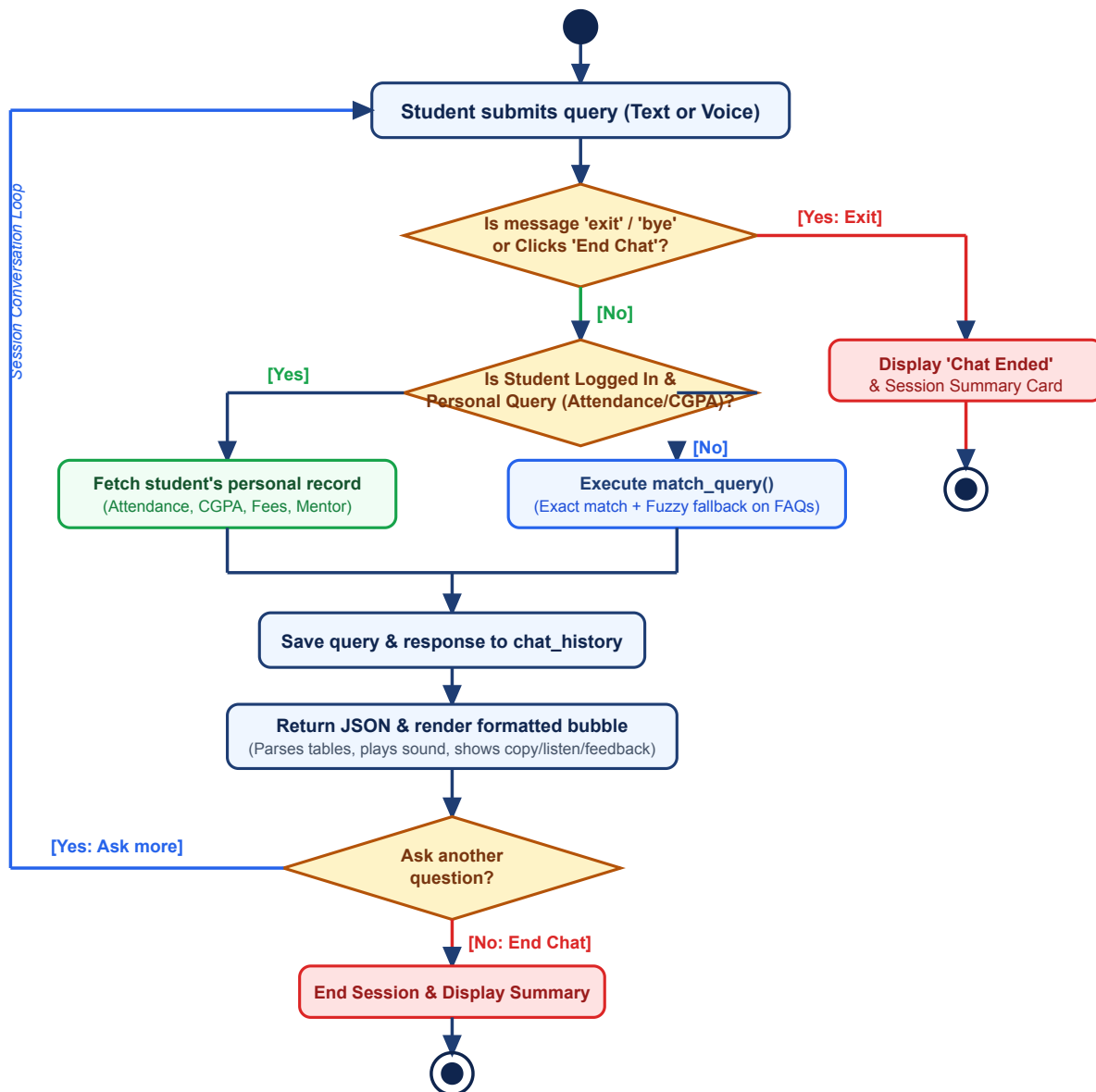


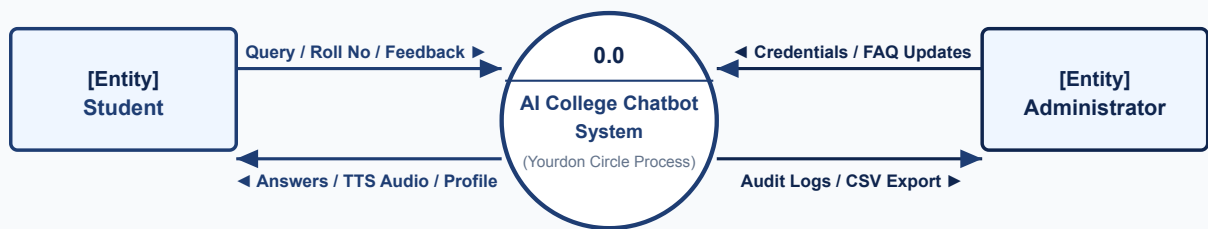
Fig 5: Activity Diagram — decision flow from receiving a message to displaying the response, showing the controlled session loop and explicit End state.

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6. Data Flow Diagram (DFD Level 0 Context & Level 1 Detailed)

DFD Level 0: Context Diagram



DFD Level 1: Process Decomposition (Circles)

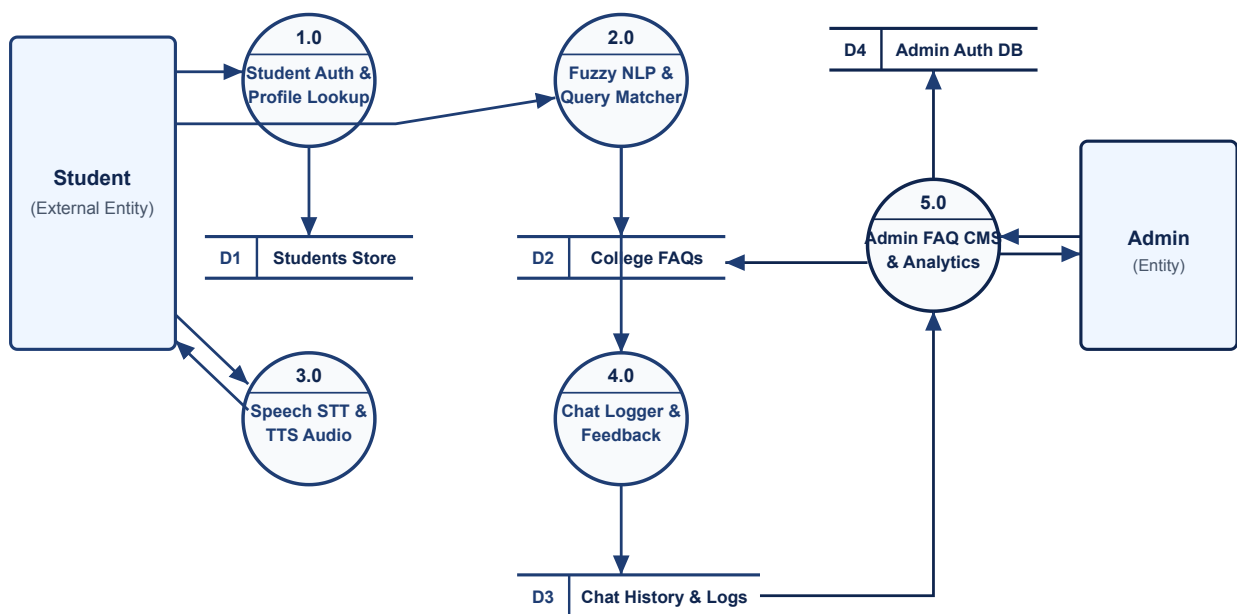
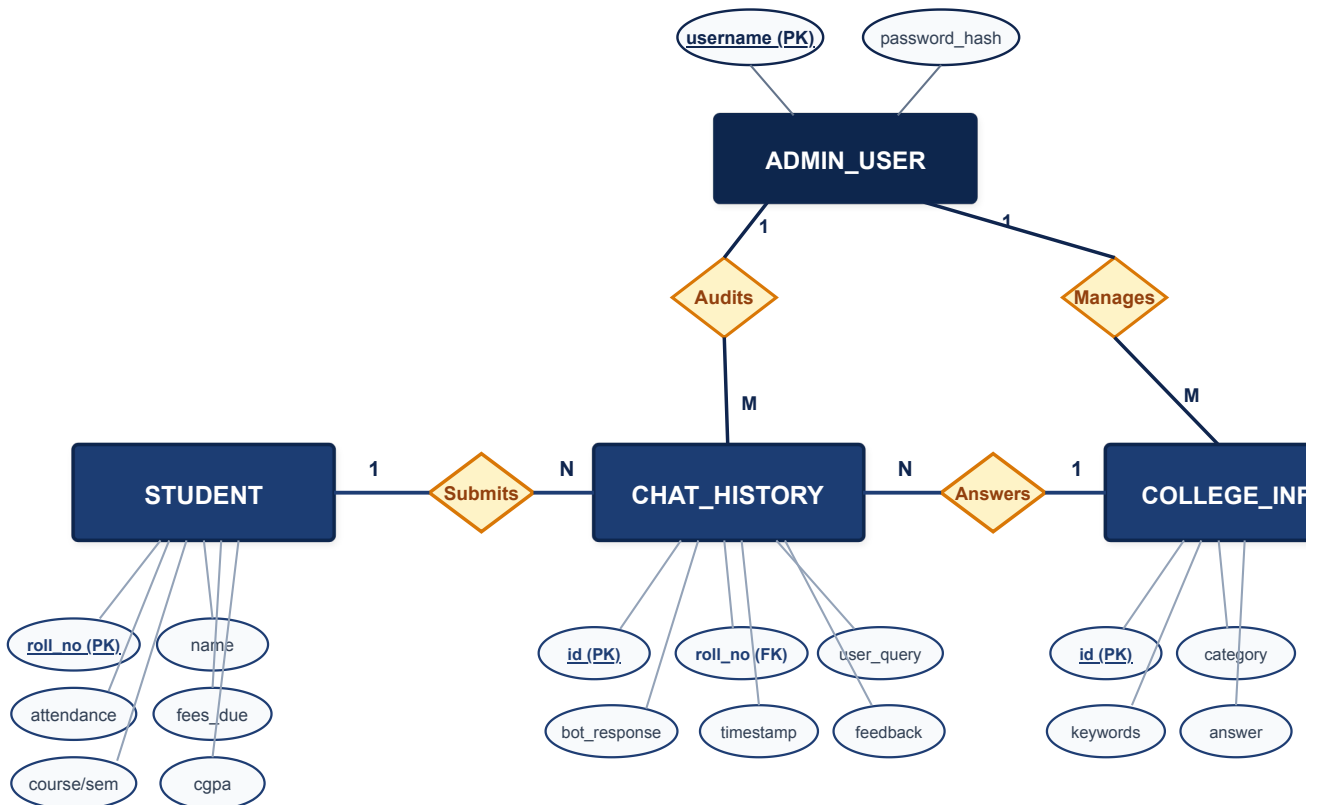


Fig 6: Data Flow Diagram (DFD) — Level 0 Context & Level 1 Detailed Decomposition using standard circle process symbols and solid data flow arrows.

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7. Entity-Relationship Diagram (ERD — Database Architecture)



✓ Standard Entity-Relationship Diagram Flow Summary:

- **Left-to-Right Primary Data Flow:** STUDENT [1] submits [N] CHAT_HISTORY, which answers [N:1] from COLLEGE_INFO.
- **Top-to-Bottom Admin Control Flow:** ADMIN_USER [1] audits [M] CHAT_HISTORY and manages [M] COLLEGE_INFO FAQs.
- **Relational Integrity:** Primary Keys underlined (<u>roll_no</u>, <u>id</u>, <u>username</u>); Foreign Key marked (roll_no FK).
- Zero line crossings; clean orthogonal layout; explicit relationship diamonds with standard cardinality markers.

Fig 7: Entity-Relationship Diagram (ERD) — structured left-to-right & top-to-bottom relational flow, diamond relationships, cardinalities, and clean non-crossing attributes.

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8. Deployment Diagram (Hosting Topology & Artifact Architecture)

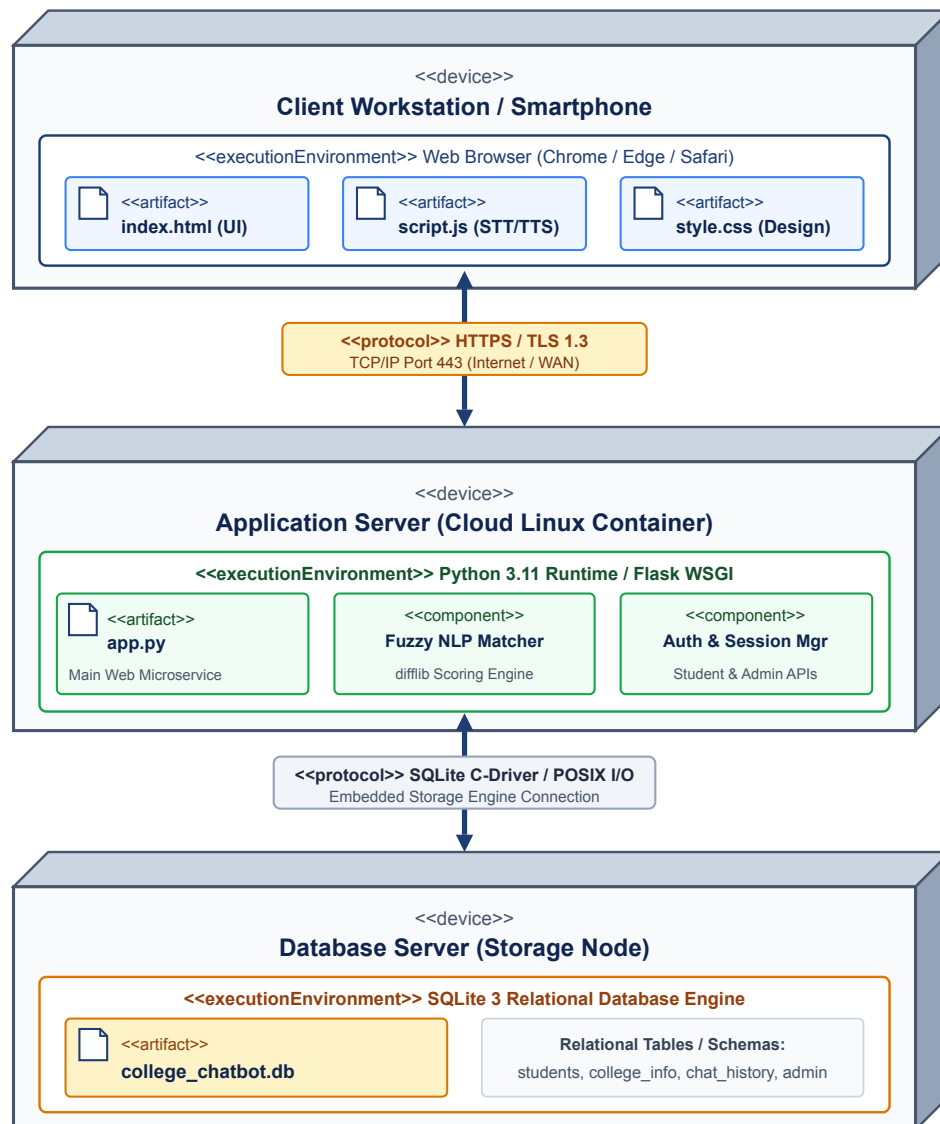


Fig 8: Deployment Diagram — textbook 3-Tier UML deployment architecture with 3D nodes (Client, Application Server, Database Server), execution environments, artifacts, and communication paths.